

**Department of Computer Science and Engineering
BRAC University**

Set A

Examination: Midterm
Duration: 1 hour 30 minutes

Semester: Fall 2025
Full Marks: 30

CSE 423: Computer Graphics

Name:	ID:	Section:
-------	-----	----------

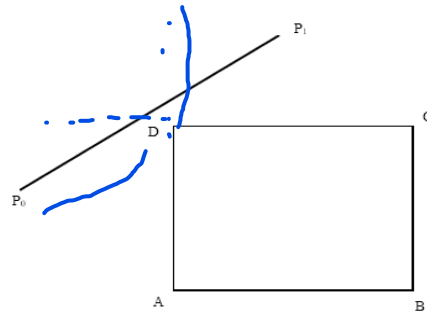
*Answer the following questions. Understanding the questions is part of the examination.
The numbers on the right margin indicate marks.*

Question 1 [CO1]

- a. A digital mapping system utilizes the **Digital Differential Analyzer (DDA)** algorithm to render objects by plotting **discrete** pixel points along a line. When the view is zoomed in, the objects appear jagged along their diagonal edges. Without modifying the hardware, propose any method that could be applied to minimize this jagged visual effect. Now, if the system used a continuous beam drawing method instead, would the DDA algorithm be necessary to generate these lines? In two or three sentences, explain why or why not. 2+2
- b. Suppose, for your CSE423 Lab Assignment 1, you need to draw a slanted line with starting point (220, 111) and ending point (111, -111). However, instead of using `GL_LINES`, you need to draw the line using the Midpoint Line (MPL) Drawing algorithm. Now, using the MPL Algorithm, **determine** the **first six** pixels (including the starting pixel) of this line. Show the necessary calculations in a table. 6

Question 2 [CO2]

- a. You are given a rectangular clipping window **ABCD** and a line segment **P_0P_1** as shown in the figure. The line segment **P_0P_1** is clearly **outside** the clipping window **ABCD**. However, the **Cohen–Sutherland line clipping algorithm** cannot **directly reject** this line. Instead, it subdivides the line before rejection. 4



Do you agree with this statement? **Explain** and **justify** your answer in two or three sentences.

- b. Suppose a rectangular clipping window is defined as $(-120, -100)$ and $(120, 100)$. A line segment with endpoints $(15, 90)$ and $(100, -130)$ needs to be drawn on this window. Using a **parametric** line clipping algorithm, **compute** the clipped segment of this line that can be visible in the window. Show the necessary calculation in a table with the updated values of t_E and t_L . Also **include** a simple drawing of the scenario showing the intersection points. (The drawing does not need to be exact; only a rough estimate is enough.)

Question 3 [CO3]

Kaguya and Shirogane are the two top elite students of Shuchiin Academy. While studying affine transformations in R^3 , Kaguya presented the following three equations for a combined transformation:

$$\begin{aligned}x' &= 10 + 7z - ax \\y' &= -9y - 5 - 3z \\z' &= cz\end{aligned}$$

Kaguya stated that this single composite operation could be decomposed into three standard transformation matrices. He also mentioned that the matrix, which has missing values a and c , represents a **uniform scaling**. One of these three transformations is not a **linear transformation**. He named it M_2 .

- a. **Identify** the values of a and c . Then, **identify** the three standard transformations referenced in the passage and **show** their **4x4** homogeneous matrices. 1+3
- b. Now, Shirogane needs to calculate the final position of a point after applying the three operations in a specific sequence: 2+4
- Previously mentioned **Uniform Scaling** is applied first.
 - Then, the transformation M_2 is applied
 - The **remaining transformation** is applied last.

Find out the composite transformation matrix formed by these 3 operations in the specific order. Then, apply this composite transformation to the point $P(2, 5, 10)$ with respect to another point $(-2, 4, -1)$. **Determine** the final coordinates of the transformed point, P' .

***** The End *****